

Mobile App Development with Thunkable



BIS605

Software Development Technologies for Digital Business

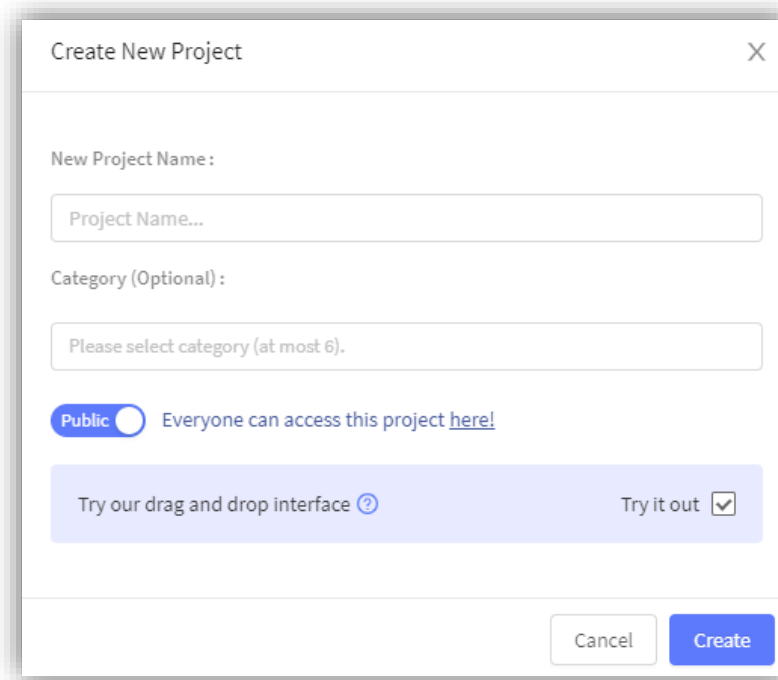
Asst. Prof. Dr. Worarat Krathu

Thunkable

- A low-code platform for building mobile apps (without coding)
 - drag and drop components
- Works both Android and iOS devices

Getting Start

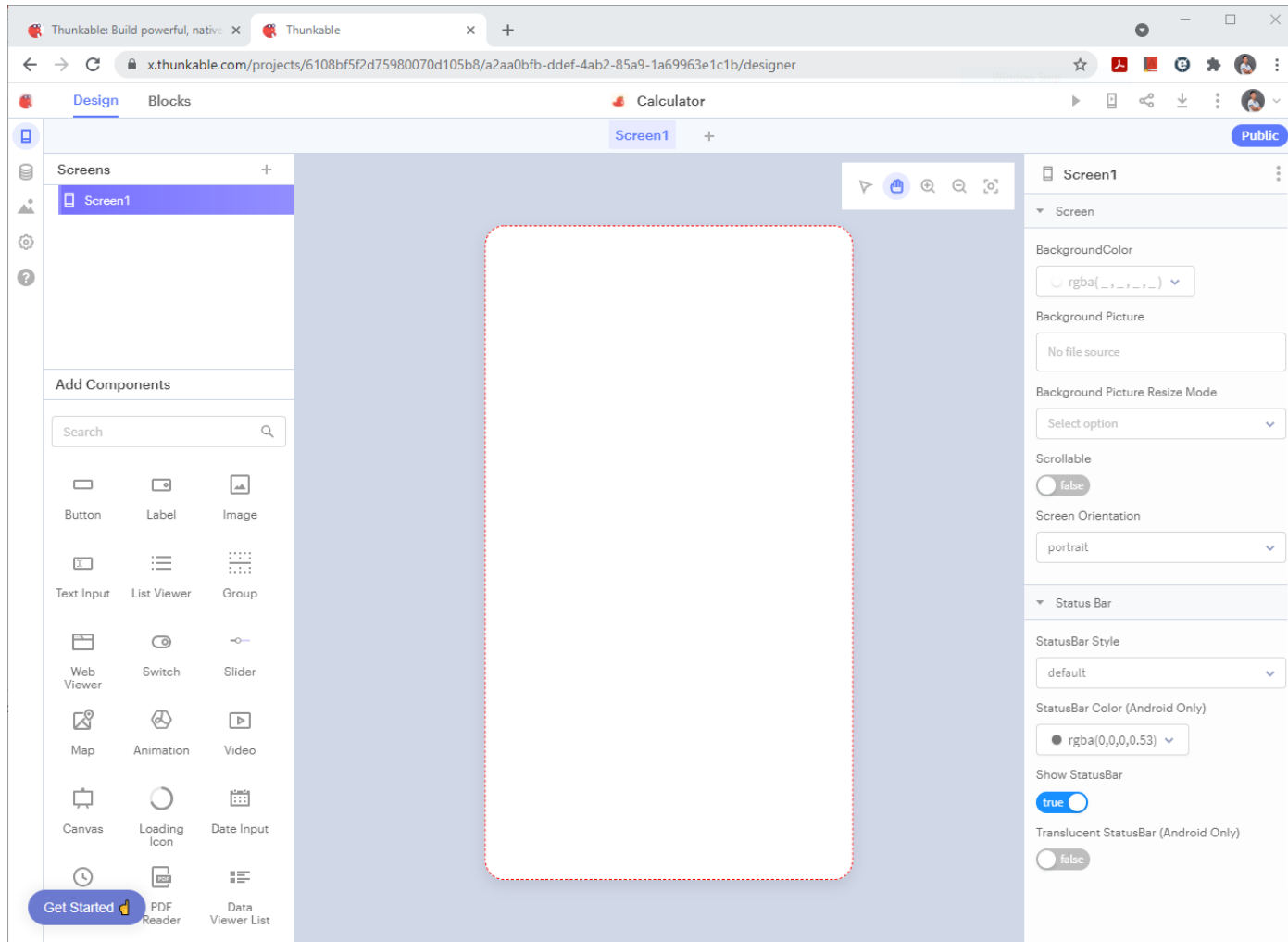
- Sign up
- Create New Project



The screenshot shows a 'Create New Project' dialog box with a close button (X) in the top right corner. It contains the following fields and options:

- New Project Name :** A text input field with the placeholder text 'Project Name...'.
- Category (Optional) :** A text input field with the placeholder text 'Please select category (at most 6)'.
- Public** (radio button selected): A label with a blue circle containing a white dot, followed by the text 'Everyone can access this project [here!](#)'.
- Try our drag and drop interface ?** (checkbox): A label with a question mark icon, followed by a checked checkbox.
- Try it out** (checkbox): A label with a checked checkbox.
- Buttons:** 'Cancel' and 'Create' buttons at the bottom right.

Getting Start



BMI Calculator Project

- This tutorial will create an application to calculate BMI
- The application will take 2 inputs
 - Weight (kg.)
 - Height (cm.)
- The application will calculate BMI from the given inputs and display as number and picture icon
- The logic of calculation
 - $BMI < 18.5 \rightarrow$ Thin
 - $18.5 \leq BMI \leq 24.9 \rightarrow$ Healthy
 - $BMI > 24.9 \rightarrow$ Fat

Please input your weight and height

Weight (kg.)
60

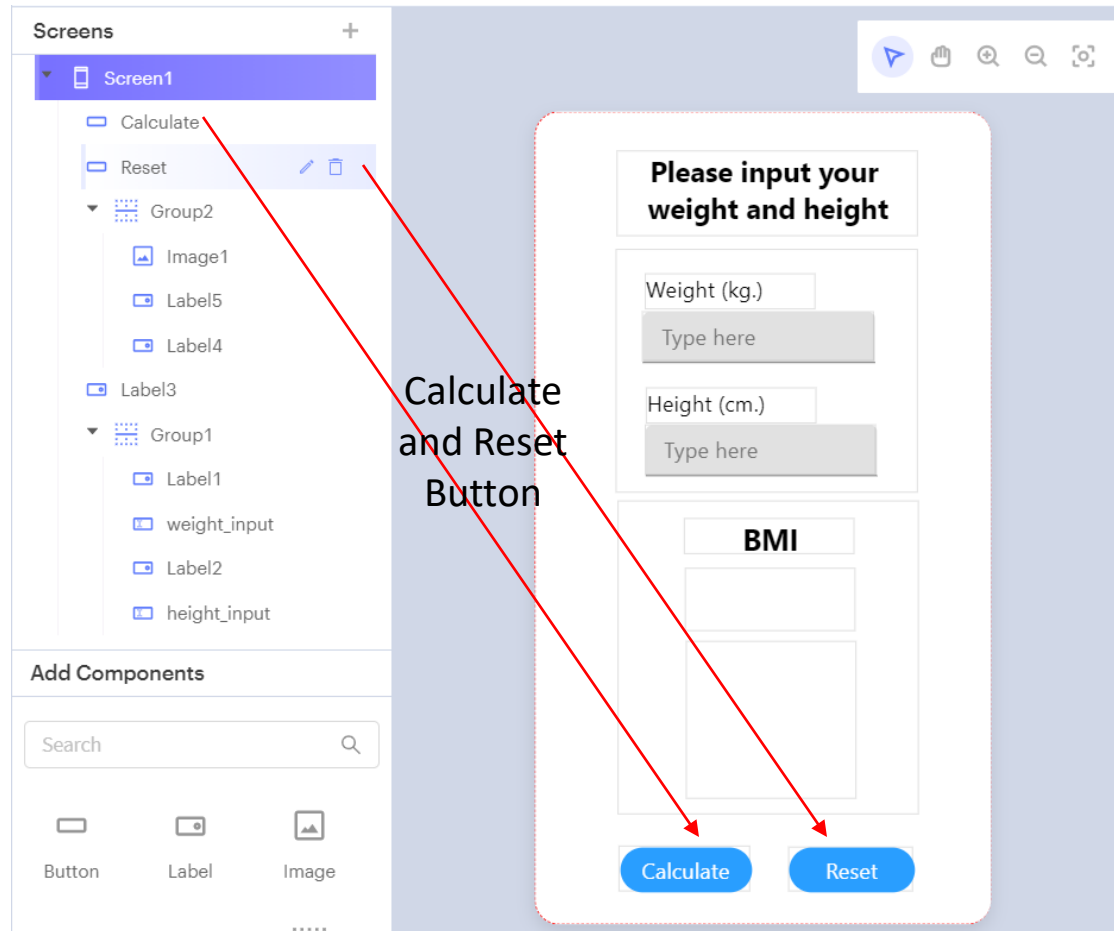
Height (cm.)
155

BMI
24.97

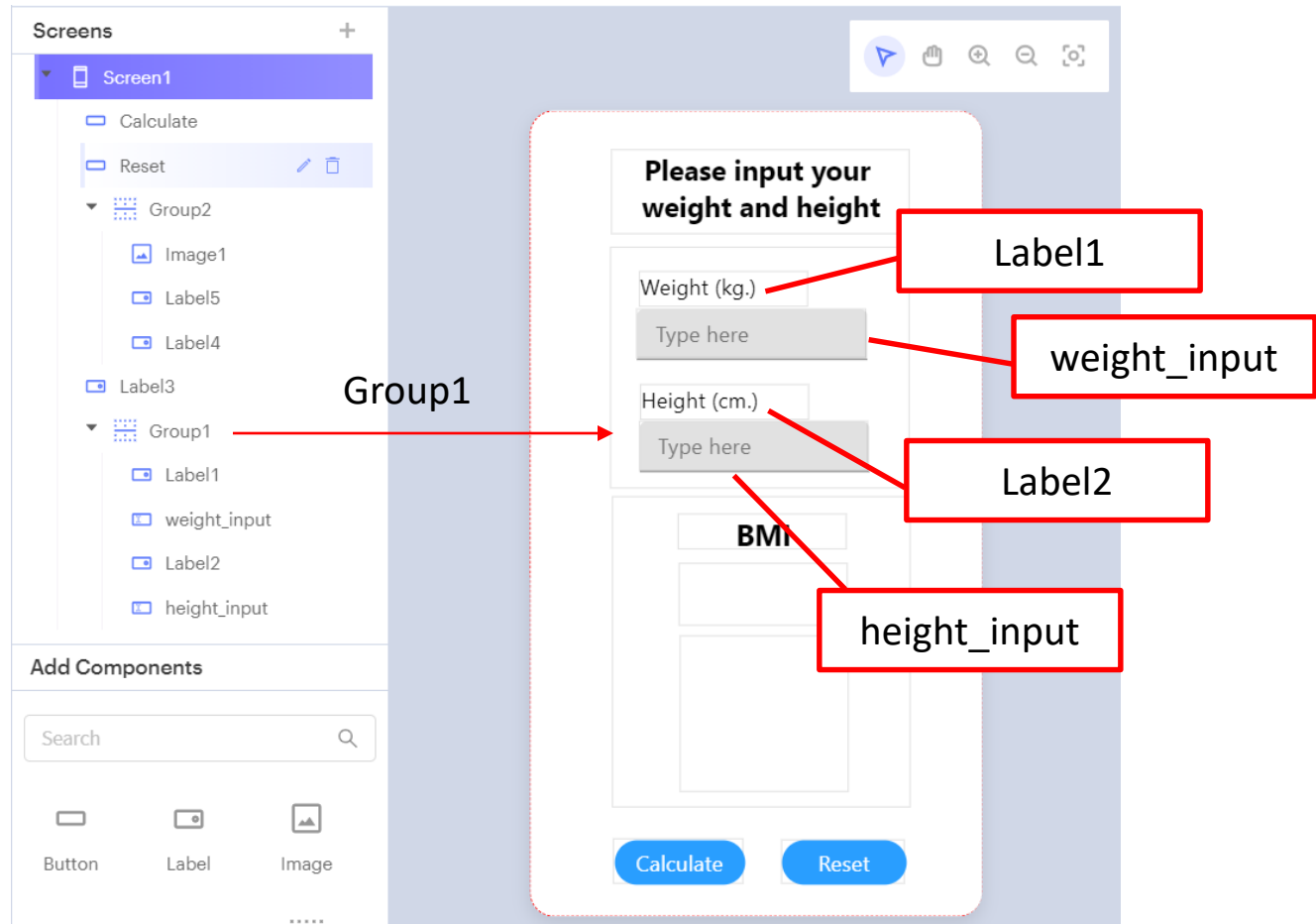


Calculate Reset

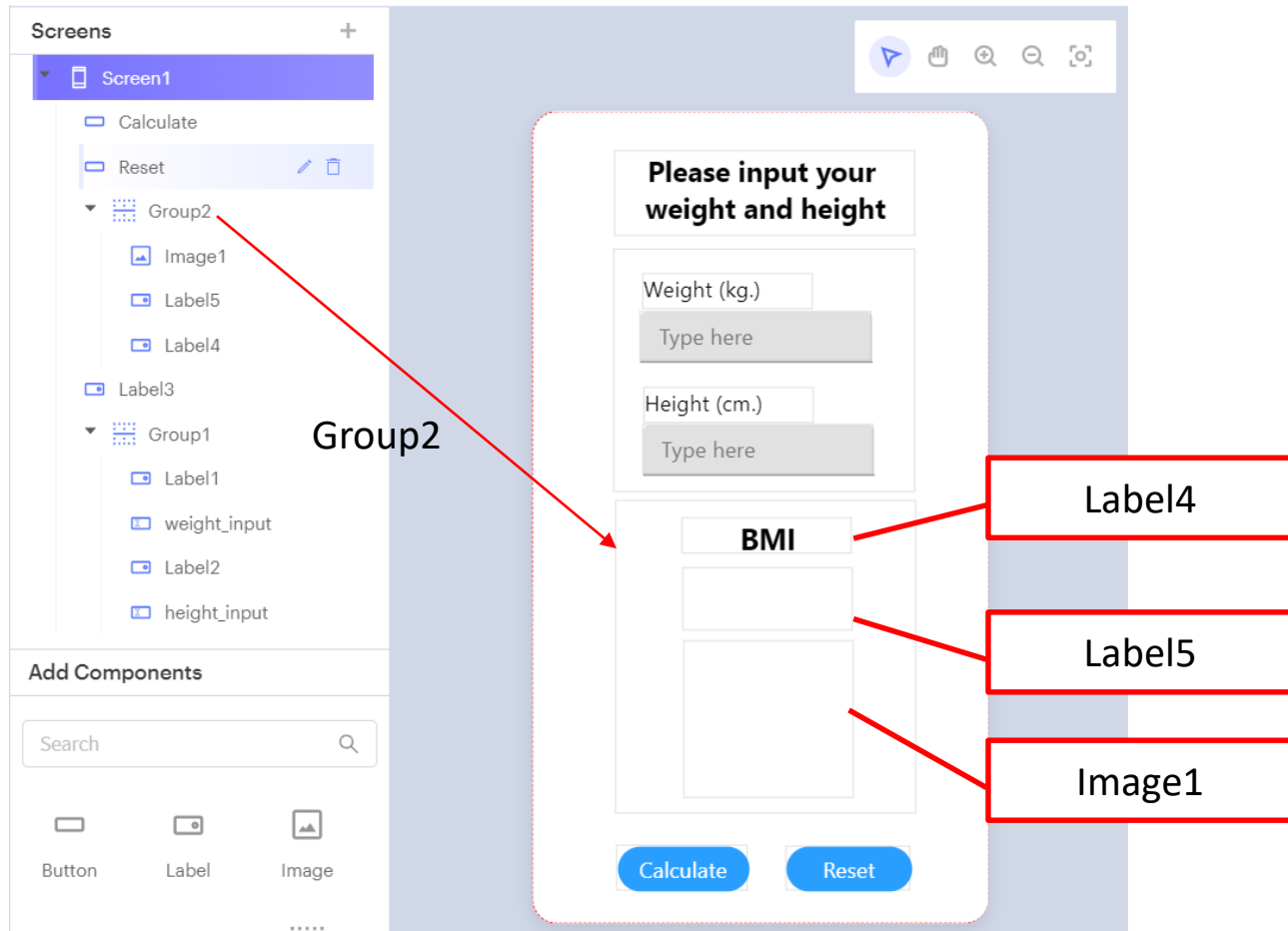
Create a Screen



Create a Screen

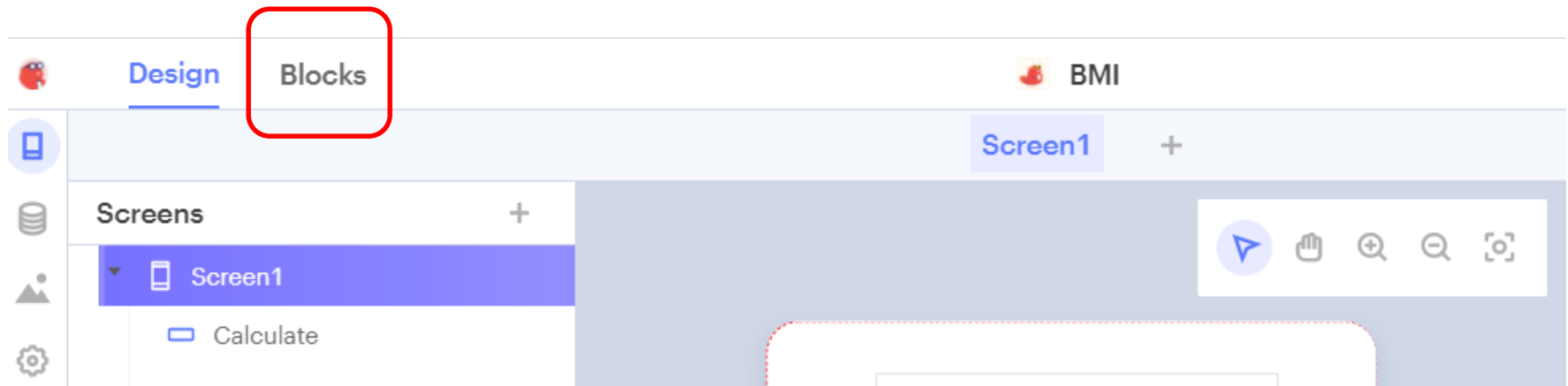


Create a Screen



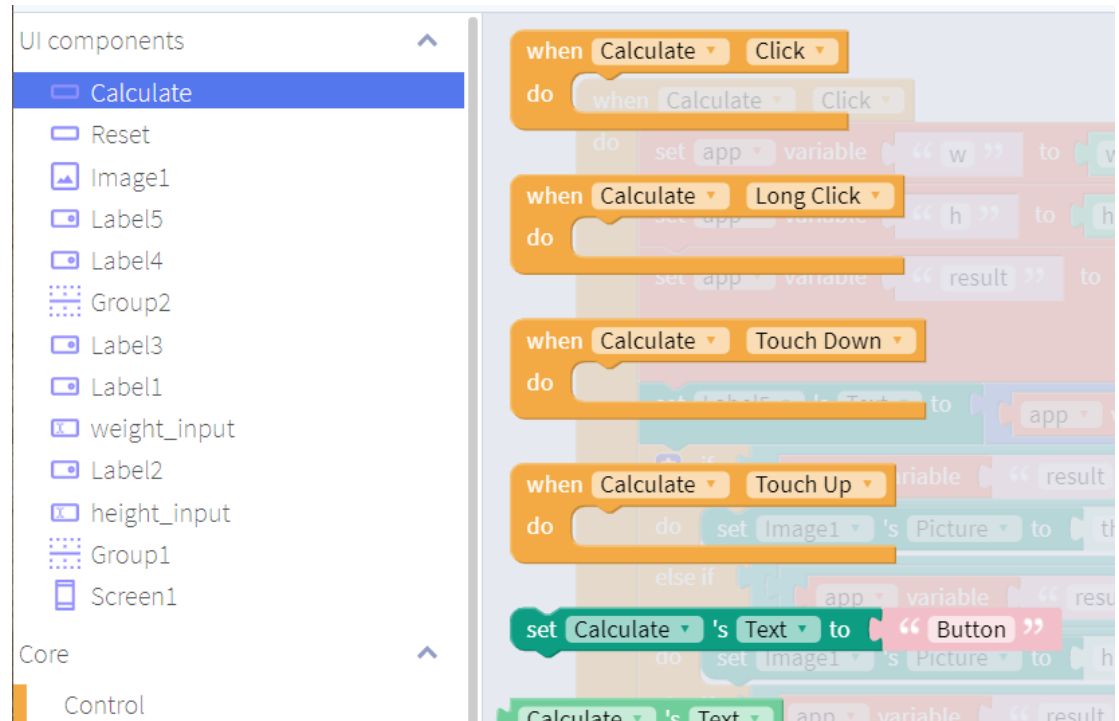
Add Logic Blocks

- Go to Blocks tab



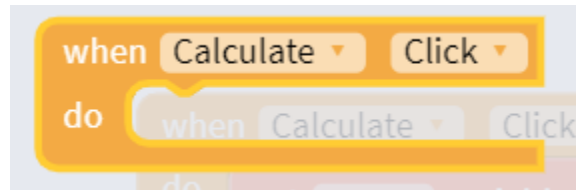
Add Logic Blocks

- Add logic to the Calculate button
- Click at Calculate (in UI components), you will see lots of actions that can add



Add Logic Blocks

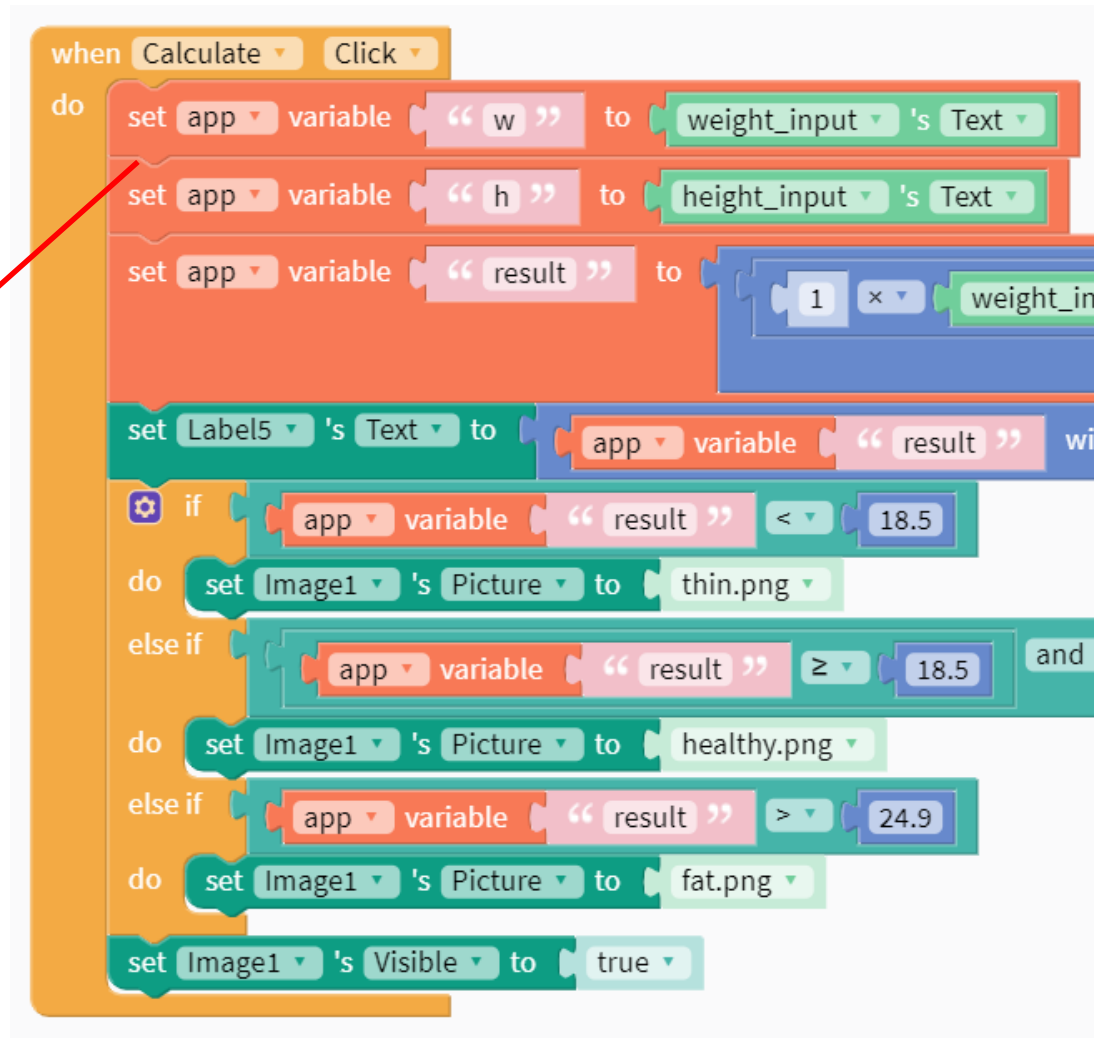
- Click at Calculate (in UI components), and add the block click block



- Add the other logic blocks as following slides

Add Logic Blocks

Go to variable blocks, add set app variable and give its name as “w” and “h” to represent the weight and height from the weight_input and height_input respectively (click at weight_input and height_input, you will find the green block to get the text of the input)

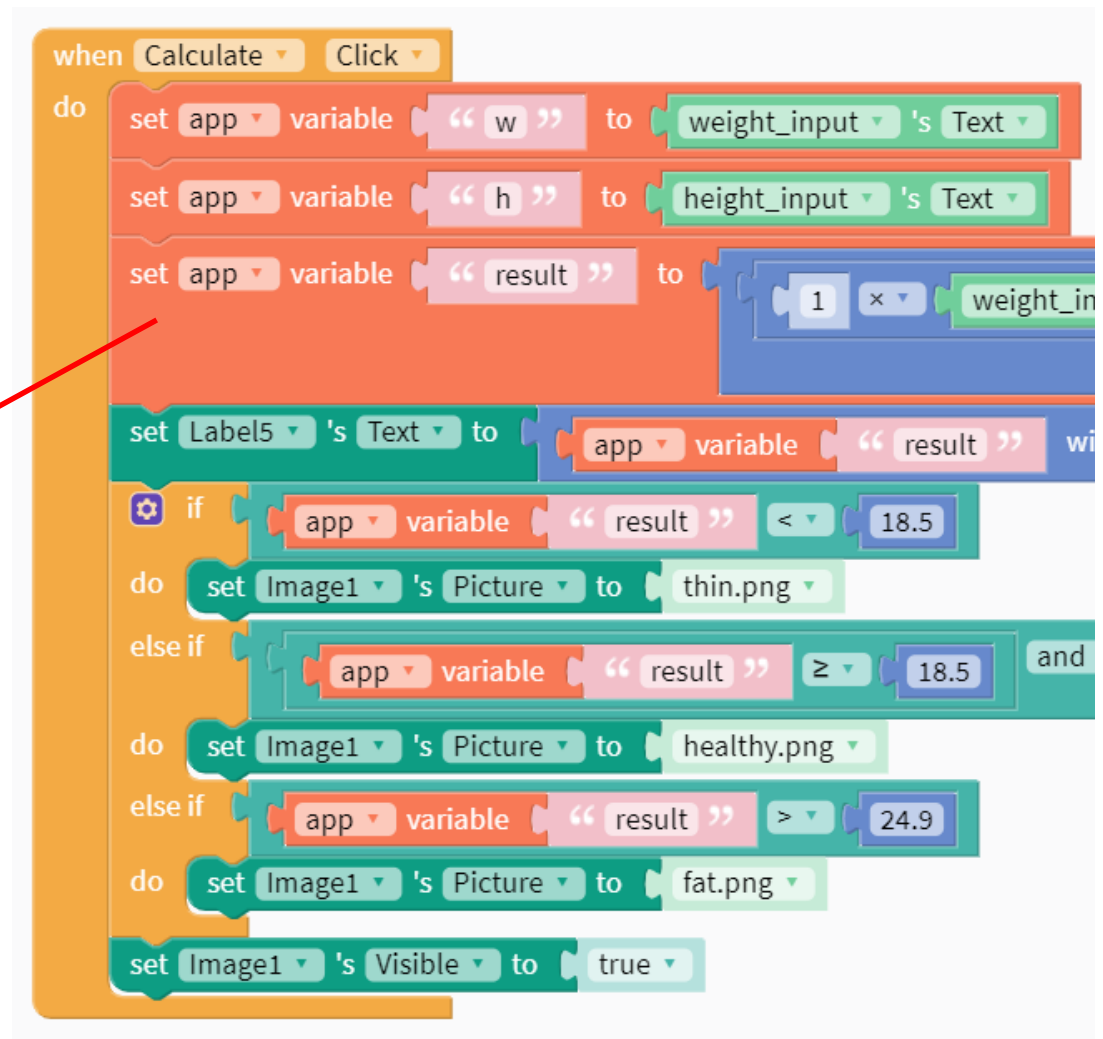


```
when Calculate button clicked
do
  set app variable "w" to weight_input's Text
  set app variable "h" to height_input's Text
  set app variable "result" to 1 * weight_input's Text / height_input's Text ^ 2
  set Label5's Text to app variable "result"
  if app variable "result" < 18.5
  do
    set Image1's Picture to thin.png
  else if app variable "result" >= 18.5 and app variable "result" < 24.9
  do
    set Image1's Picture to healthy.png
  else if app variable "result" >= 24.9
  do
    set Image1's Picture to fat.png
  set Image1's Visible to true
```

The image shows a Scratch script for calculating BMI. It starts with a 'when Calculate button clicked' event. The script then sets three app variables: 'w' (weight), 'h' (height), and 'result' (BMI calculation). The 'result' variable is calculated as $1 \times \text{weight_input} / \text{height_input}^2$. The script then sets the text of Label5 to the value of the 'result' variable. It follows with a series of if-else-if statements to determine the BMI category: if 'result' is less than 18.5, set Image1's Picture to 'thin.png'; if 'result' is greater than or equal to 18.5 and less than 24.9, set Image1's Picture to 'healthy.png'; if 'result' is greater than or equal to 24.9, set Image1's Picture to 'fat.png'. Finally, it sets Image1's Visible property to true.

Add Logic Blocks

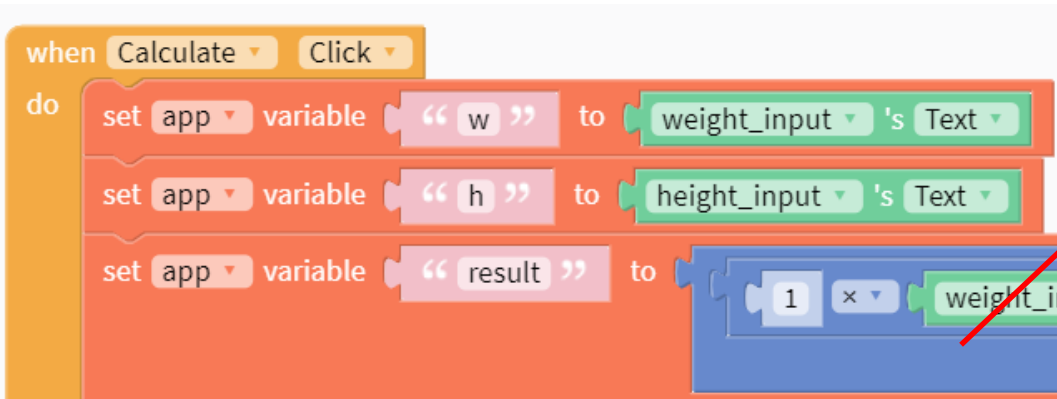
Go to variable blocks, add set app variable and give its name as “result” and set it to a math block where you will put the calculation in this math block.



```
when Calculate Click
do
  set app variable "w" to weight_input's Text
  set app variable "h" to height_input's Text
  set app variable "result" to 1 × weight_in
  set Label5's Text to app variable "result" wi
  if app variable "result" < 18.5
  do
    set Image1's Picture to thin.png
  else if app variable "result" ≥ 18.5 and
  do
    set Image1's Picture to healthy.png
  else if app variable "result" > 24.9
  do
    set Image1's Picture to fat.png
  set Image1's Visible to true
```

The image shows a Scratch script for calculating BMI. It starts with a 'when Calculate Click' event. The script then sets three app variables: 'w' to the text of 'weight_input', 'h' to the text of 'height_input', and 'result' to the product of 1 and 'weight_in'. The 'result' variable is then used to set the text of 'Label5'. The script then uses a series of 'if' and 'else if' blocks to set the 'Picture' of 'Image1' based on the value of 'result'. Finally, it sets 'Image1's Visible' to 'true'.

Add Logic Blocks



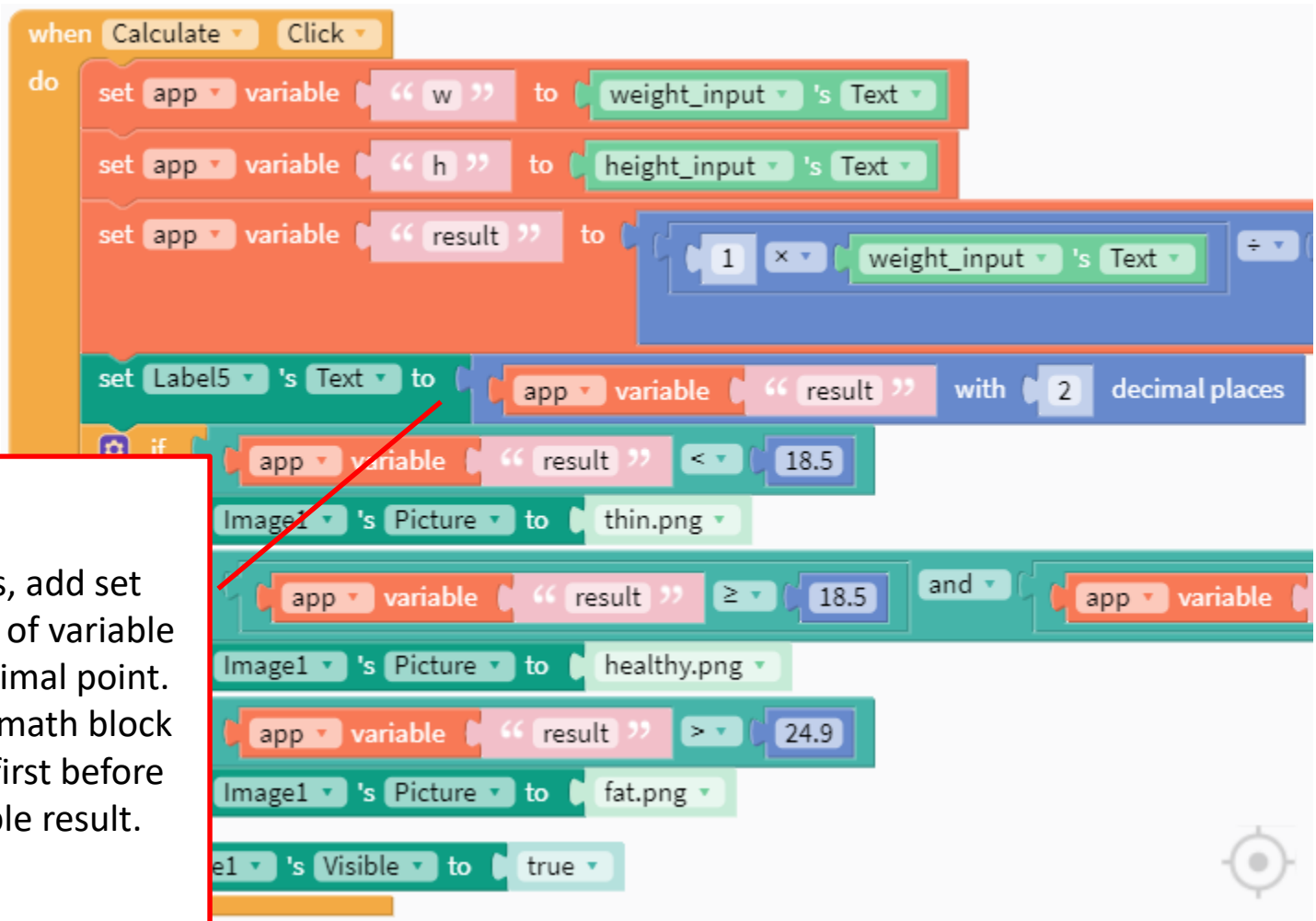
In the math block, set the calculation as shown below.



In the math block, build the calculation for BMI.
$$\text{weight} / (\text{height} / 100)^2$$

Note that we multiply 1 at the weight and weight inputs for converting the type of data from Text to Number.

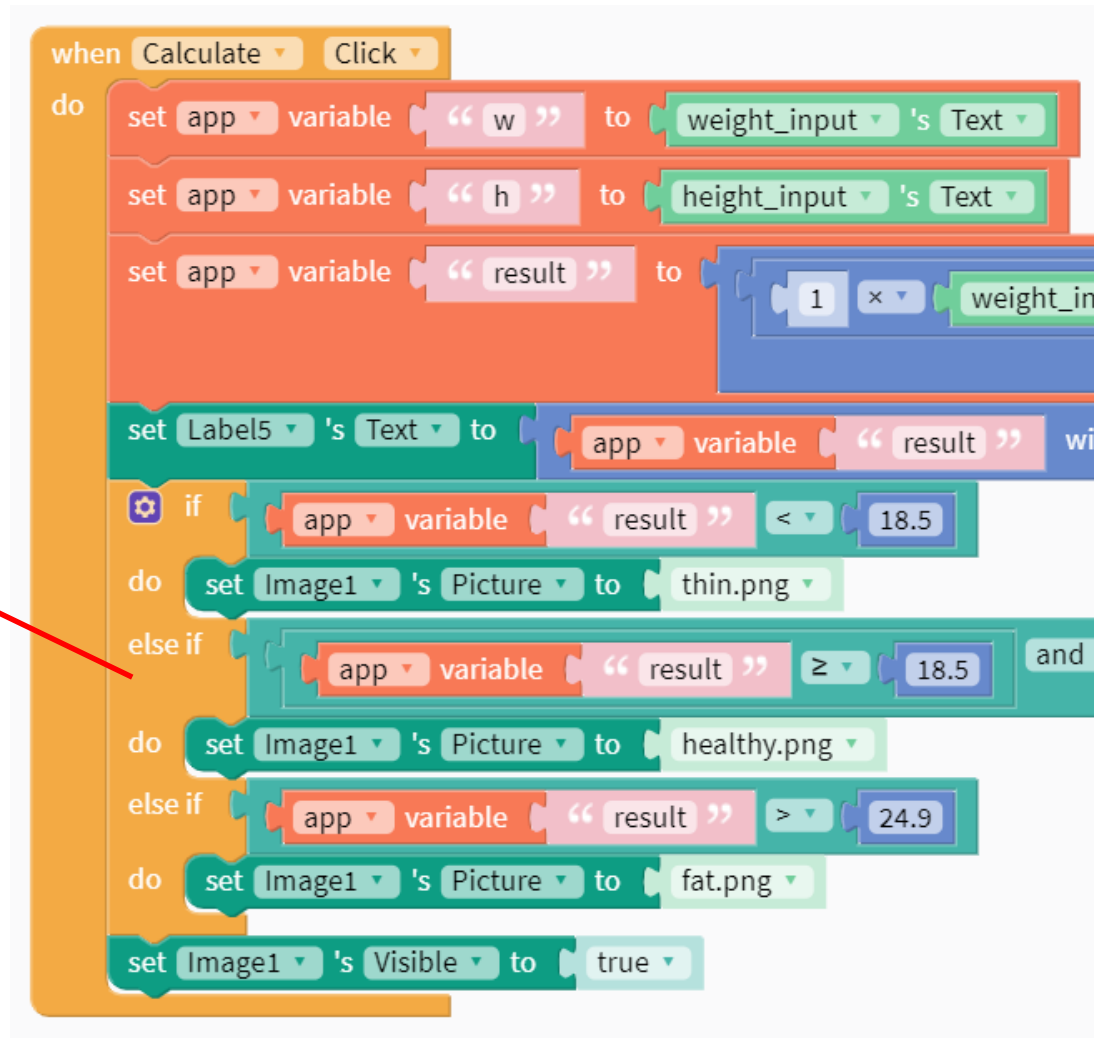
Add Logic Blocks



Go to logic blocks, add set Label5 to the value of variable "result" with 2 decimal point. (You must add the math block of decimal places first before adding the variable result.)

Add Logic Blocks

Go to control blocks, add the If block. Click at the gear icon, add 2 else if blocks under the if block. You will get the if block with two else if blocks as shown in the picture.



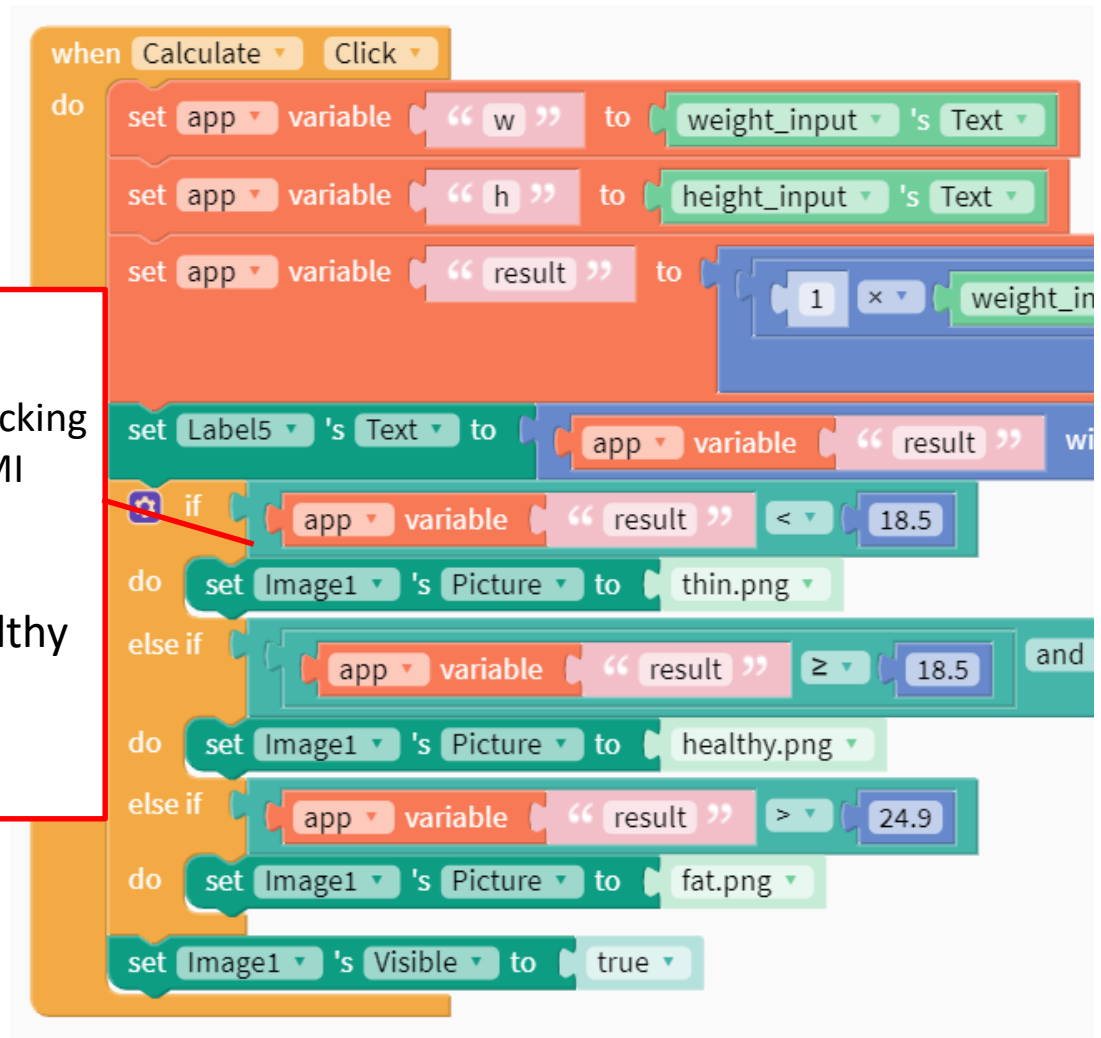
Add Logic Blocks

In the If block, we will add condition checking the value of “result” according to the BMI evaluation:

BMI < 18.5 -> Thin

18.5 ≤ BMI ≤ 24.9 -> Healthy

BMI > 24.9 -> Fat



Add Logic Blocks

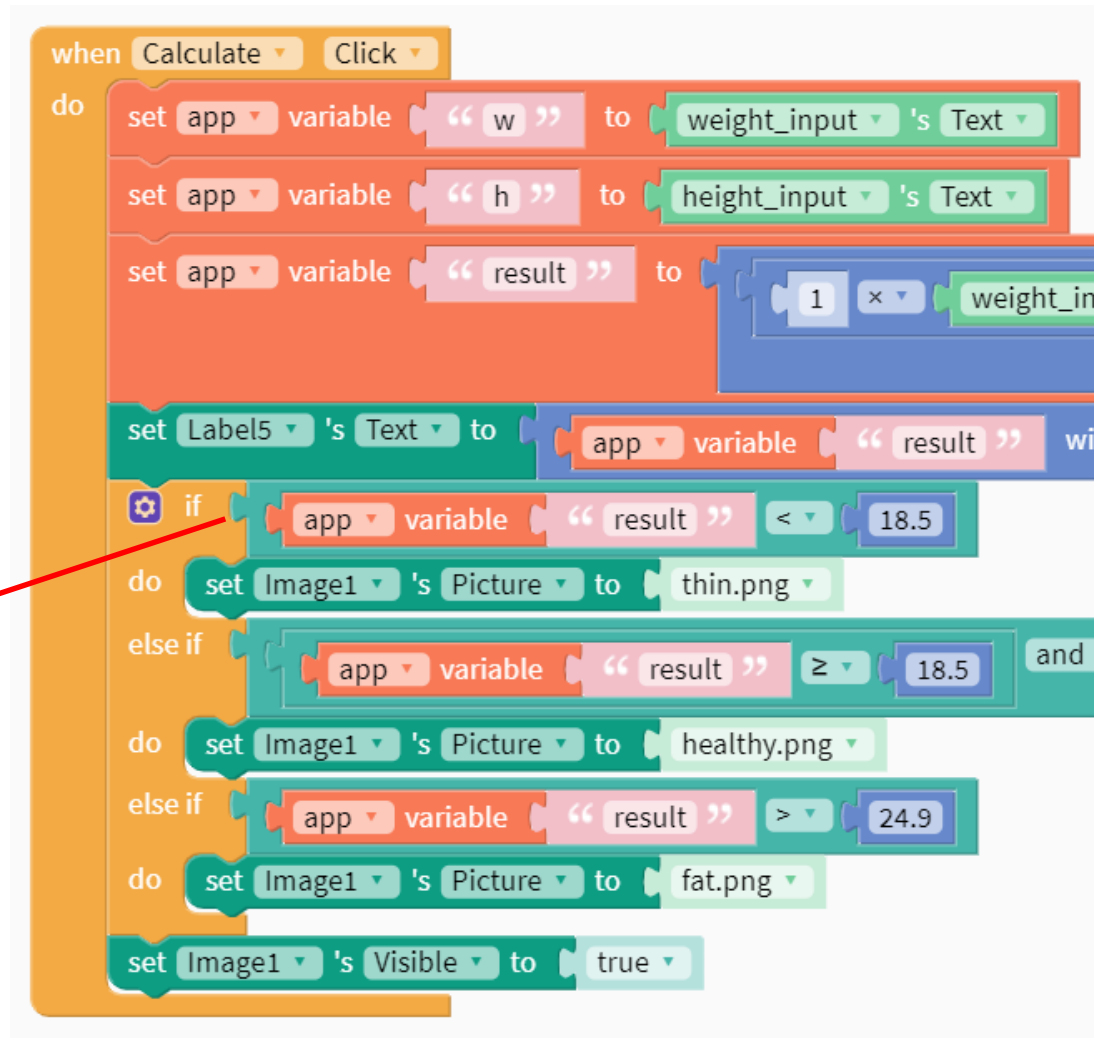
The first condition:

If

the “result” < 18.5

Do

Set Image1 to thin man image
(you can use any images as you want)



Add Logic Blocks

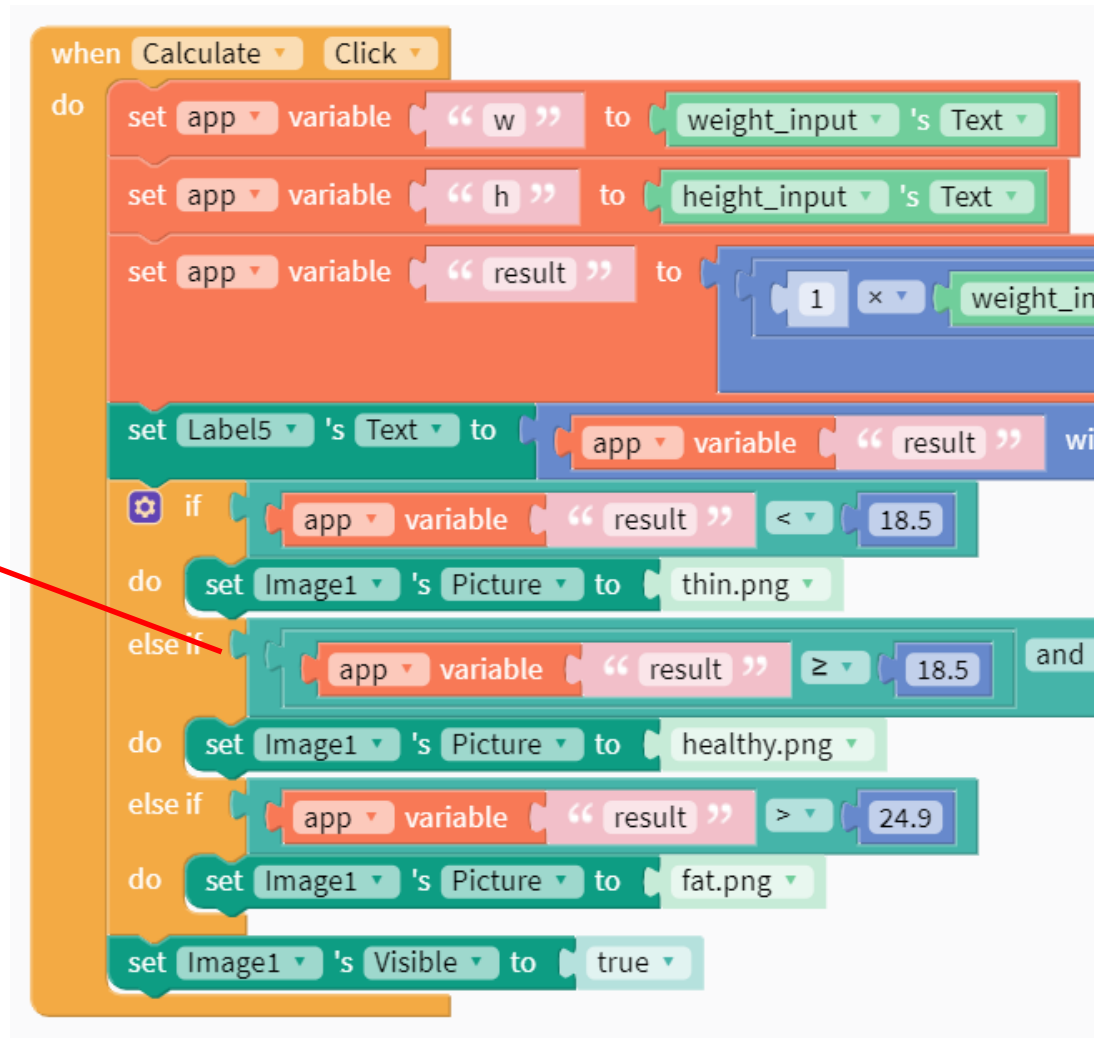
The second condition:

Else If

the “result” ≥ 18.5 and “result” ≤ 24.9

Do

Set Image1 to healthy man image



Add Logic Blocks

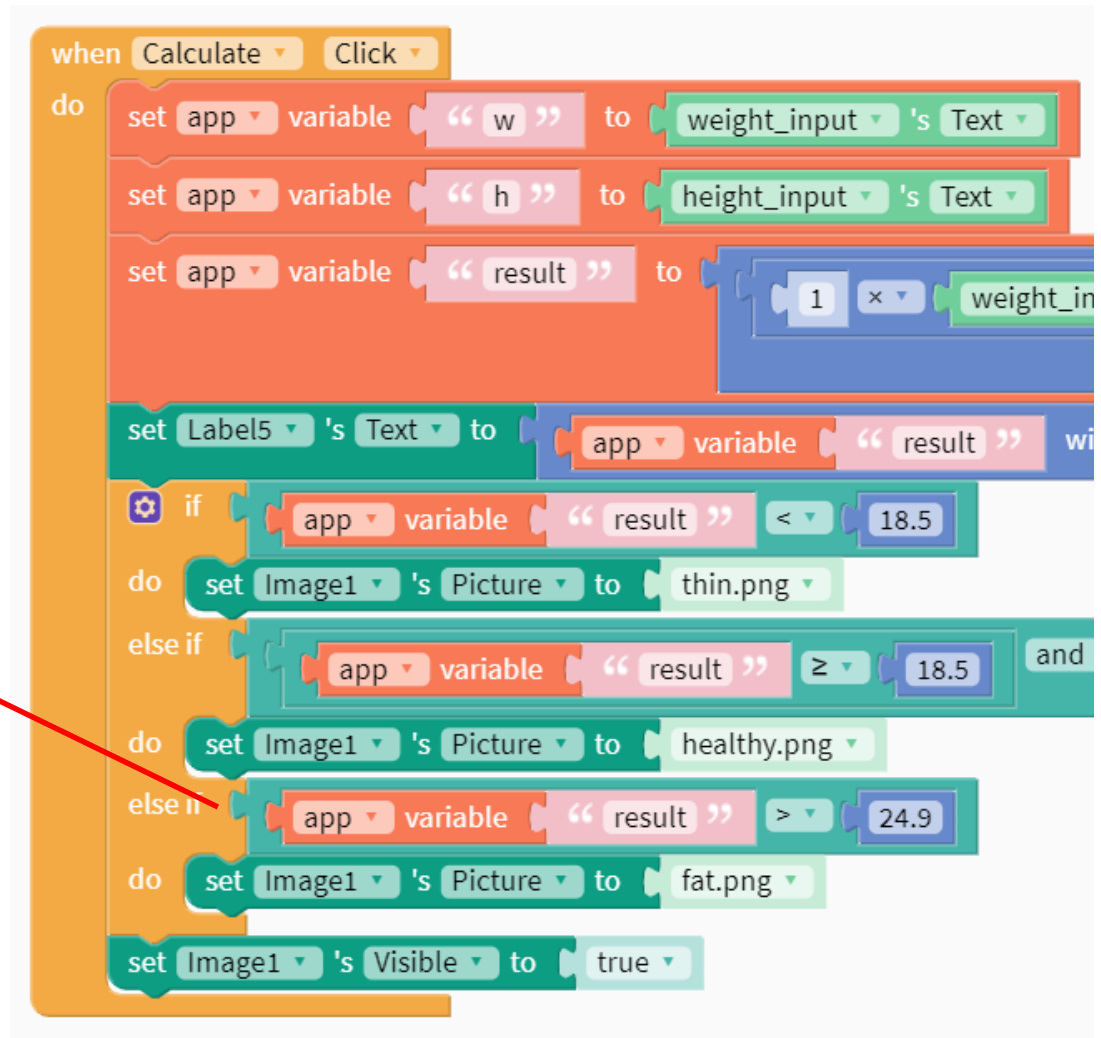
The third condition:

Else If

the “result” ≥ 24.9

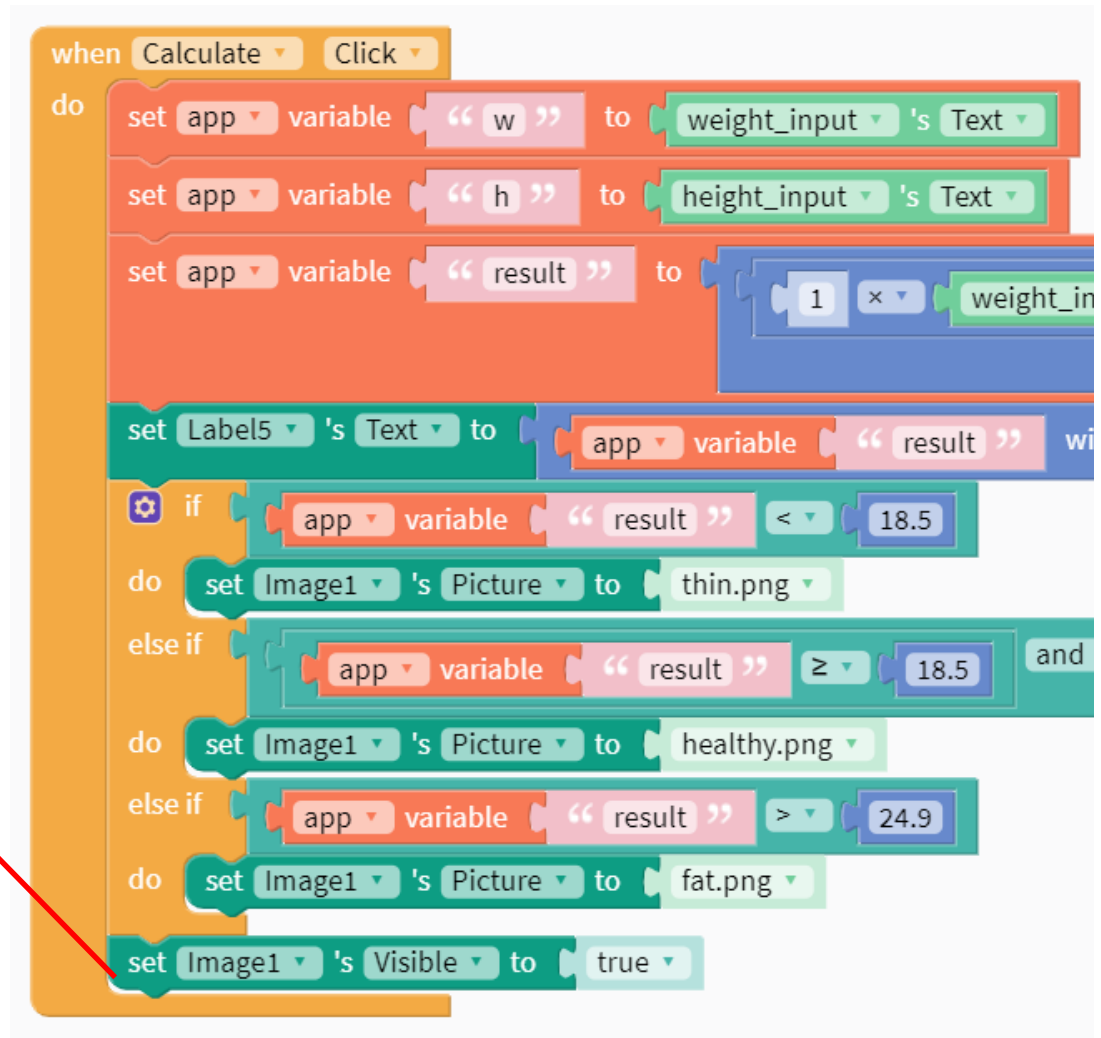
Do

Set Image1 to fat man image



Add Logic Blocks

After the result is evaluate, the image is set to visible

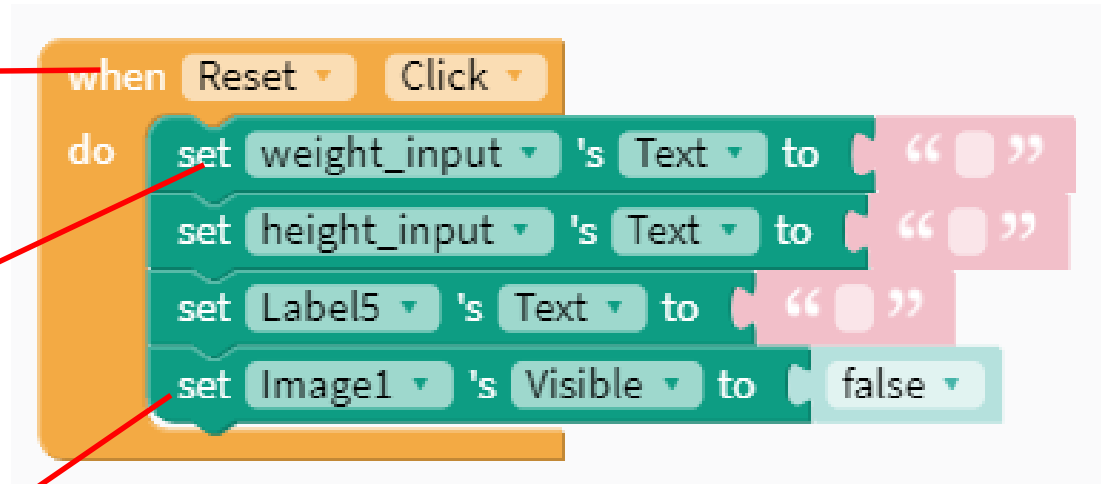


Add Logic Blocks

Add the Click block of Reset button

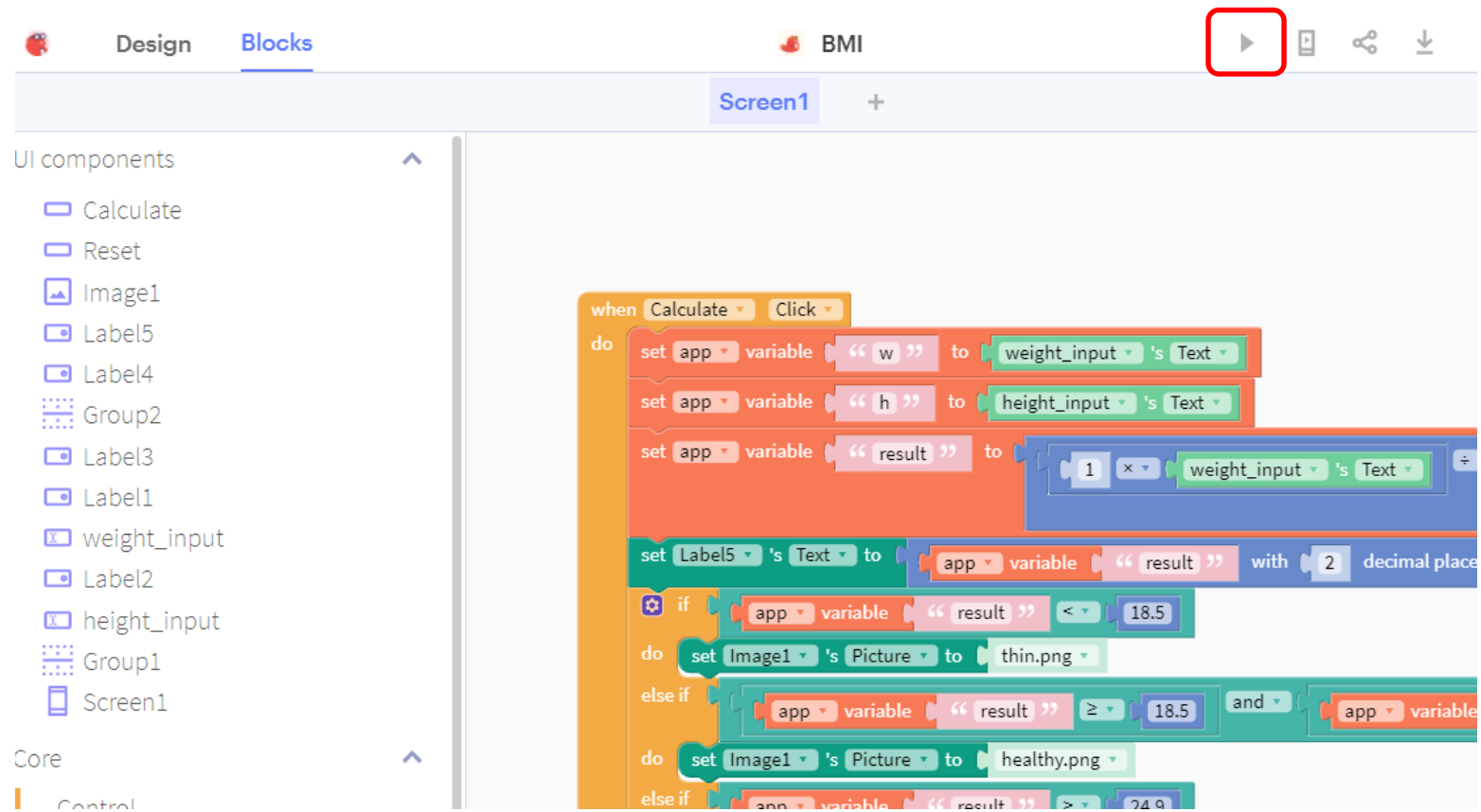
Set the weight_input, height_inpt and Label5 to "" (empty text)

Set Image1 's visible to false



Test the App

- Click play button



Test the App

Please input your weight and height

Weight (kg.)
60

Height (cm.)
155

BMI
24.97



Calculate Reset

Exercise 12.1

- Create your own mobile app using thinkable. Submit as a shared project link.